

PROJECT AND TEAM INFORMATION

Project Title

CHESSVERSE

Student / Team Information

<i>Team Name:</i> <i>Team #</i>	<i>Checkmates</i> <i>DSCPP-III-2026-T175</i>
Team member 1 (Team Lead) <i>(Last Name, name: student ID: email, picture):</i>	<i>Singh, Priyani –</i> <i>2510019948</i> <i>priyanis997@gmail.com</i> 
Team member 2 <i>(Last Name, name: student ID: email, picture):</i>	<i>Kotnala, Arnav –</i> <i>2510380036</i> <i>kotnalaarnav2007@gmail.com</i> 
Team member 3 <i>(Last Name, name: student ID: email, picture):</i>	<i>Gairola, Swastik –</i> <i>2510380064</i> <i>tanmaygairola69@gmail.com</i> 
Team member 4 <i>(Last Name, name: student ID: email, picture):</i>	<i>Riya –</i> <i>2510376383</i> <i>rosieriyaa@gmail.com</i> 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Chess has always been more than a game of moving pieces. It is a game of strategy. It is a game of making decisions. It is a game of trying to figure out what the other person will do next. These things made chess one of the problems that people looked at when they wanted to know if machines could act like humans. This helped start the work on Artificial Intelligence. The link between chess and AI made us want to build ChessVerse. With this project we want to learn how the rules of chess and the choices people make can be shown in a computer. We want to add an AI opponent. ChessVerse is a way to see how a simple board game can put together strategy, computer programs, networks and Artificial Intelligence all, in one app.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Today chess is played both in life and online on sites like Chess.com and Lichess. These platforms give players online multiplayer games, matchmaking, game timers, analysis, rankings and even computer opponents. Modern chess programs use algorithms to map the chessboard list all legal moves score possible positions and pick the best move. Old-school methods such, as the Minimax algorithm look ahead at moves and use evaluation tricks that mimic strategic thinking. Newer engines mix these search tricks with optimization techniques and huge computing power to play at a high level. ChessVerse is inspired by them but runs locally. Puts it into one small app. It blends chess rules with the simplicity of C language.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

- 1) ChessVerse was created step, by step. ChessVerse first made the chess piece images. Designed the chessboard. ChessVerse then added the chess moves and rules. ChessVerse also added legal move checking, game-state handling and an easy-to-use interface that makes playing smooth and fun.*
- 2) Unlike other online Chess websites, our goal is to design offline chess platform purely made with C/C++ programming.*

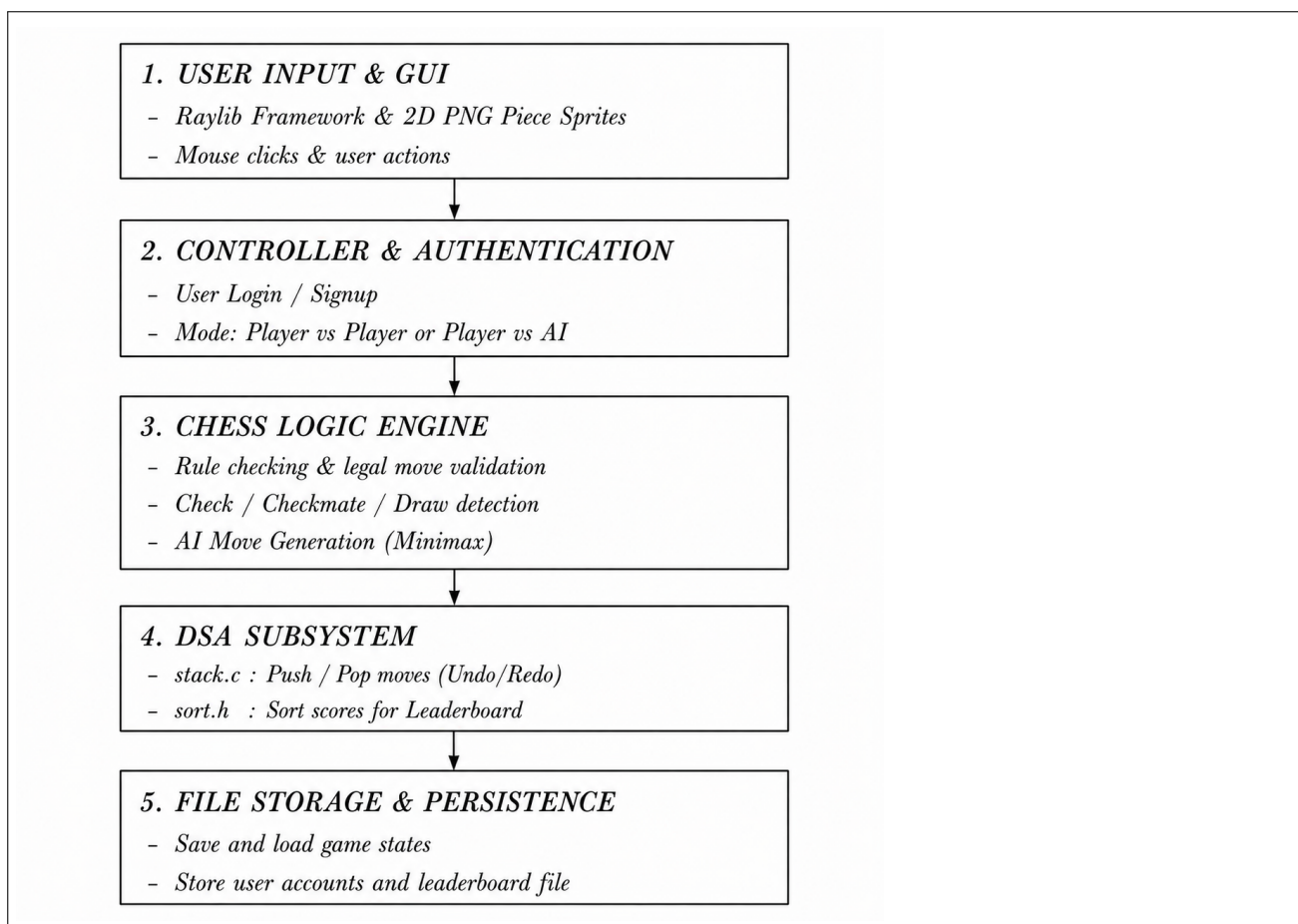
Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words.)

- 1) The project is built as an offline chess application implementing fundamentals of DSA and OOPs. It brings together the user interface, the chess engine and the AI opponent into one working system. Development begins with gathering chess-piece images and designing a chessboard. Then the chess engine is built to represent the board state check if moves are valid and manage game conditions like check, checkmate, castling and pawn promotion.
- 2) Our chess uses the Minimax algorithm with alpha-beta pruning. This approach lets the computer look ahead at moves and choose the best one making the game challenging and intelligent.
- 3) The application is meant to work locally on PC. The core logic for chess rules is written in C or C++. The user interface is purely build on C++ using its raylib library. This step-by-step testing helps make sure the chess rules are followed correctly offering a smoother game experience and the system remains reliable overall.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

- 1) Built a working ChessVerse app that works locally on PC. The app should let user play against each other on the same device or play by themselves against a computer. It needs a chessboard correct way to check moves proper chess rules and a simple interface that is easy to use.
- 2) The app should use-
Greedy 1-Ply / Random Legal Move for easy mode(depth 1)
Standard Minimax for medium mode (depth 2-3)
Minimax + Alpha-Beta Pruning for hard mode (depth 4-5)

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

Chess rules and move validation will be implemented up to the level needed for the project demonstration. The AI opponent will use the Minimax algorithm with alpha-beta pruning. The difficulty of the AI will be set manually by the user. Student IDs, emails, team number and member pictures will be added later if required.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

1. Raylib, titled "Raylib Official Documentation," from Raylib is available online at <https://www.raylib.com/games.html>.
2. Wikimedia Commons, titled "PNG Chess Pieces – Standard Transparent," from Wikimedia Commons is available online at https://commons.wikimedia.org/wiki/Category:PNG_chess_pieces/Standard_transparent.
3. Chessprogramming Wiki, titled "Chess," from Chessprogramming Wiki is available online at <https://chessprogramming.org/Chess>.
4. GeeksforGeeks, titled "Sorting Algorithms," from GeeksforGeeks can be accessed online at <https://www.geeksforgeeks.org/dsa/sorting-algorithms/>.